

PUBH 7411, Section 001

Reproducible Research in Brain Imaging Genetics
Summer 2026

Credits: 3 credits

Modality and Meeting Day(s), Time, and Place:

- This class is in-person. The lectures and lab meet daily at **9:00 - 10:30 am, 1:00 - 2:00 pm, and 2:30 - 3:30 pm** on **Monday - Friday** (July 13 - 31, 2026) at **PWB 3-150**.
- Three seminar or roundtable discussion sessions will be scheduled during lunch time (**11:30 am-12:30 pm**) on **Monday, Wednesday, and Friday** each week with **lunch provided**.
- A webinar will be held on Monday, August 31, 2026 for group presentations.

Course Site:

- All lecturing materials will be posted at canvas website <https://canvas.umn.edu/courses/565157>
- The course will publicate materials at the GitHub webpage at <https://umn-gdc.github.io/RR-BIG-Workshop/>

Instructional Team Contact Information

Instructors

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Teaching Assistant(s)

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Course Details

Course Description

An unprecedented volume of neuroimaging and genomics data become available through large-scale open data initiatives such as the Adolescent Brain Cognitive Development Study (ABCD), spurring rapid growth of brain imaging genomics studies to improve understanding of the pathophysiological processes and genetic architecture of mental health. While these large-scale repositories create new opportunities to explore the brain's interaction with the environment and integrate socioenvironmental and behavioral data with neuroimaging and genomic information, they also raise challenges in data management and analysis. The Summer Course on Reproducible Research in Brain Imaging Genetics (RR-BIG) is designed to train junior researchers in the best practices for reproducible analysis of large, openly accessible brain imaging and genomics data to learn how to estimate meaningful associations of genetic variants on brain structure and function, along with their implications for cognitive functions and broader mental health outcomes.

Course Goals and Objectives

- **Knowledge of Large Open Databases:** Learn to access and use the ABCD databases and understand analytical and statistical considerations specific to large open data.
- **Learn to Conduct Neuroimaging-Genotype Association Analysis:** Know the important scientific questions in brain imaging genomics studies, learn basic principles and analysis methods for brain imaging genomics data, and can conduct integrative analyses in R or relevant tools.
- **Reproducible Scientific Research:** Develop the best practices for ensuring transparency and reproducibility in brain research and build reproducible and interpretable analysis with good coding practice and appropriate documentation.
- **Interdisciplinary Collaboration:** Foster collaboration between early-career researchers, data scientists, and domain experts.

Prerequisites

Graduate students, postdocs, research associates, junior faculty, or other early-career researchers with research interest in brain imaging genetics data analysis and/or genetic basis of mental disorders.

Required training background in at least one of these core competencies: statistics, bioinformatics, neuroscience, or related fields.

Some experience with R or Python is desired.

Land Acknowledgement

The School of Public Health at the University of Minnesota Twin Cities is built within the traditional homelands of the Dakota people. Minnesota comes from the Dakota name for this region, Mni Sóta Maŋoŋce, which loosely translates to the land where the waters reflect the skies.

It is important to acknowledge the peoples on whose land we live, learn, and work as we seek to improve and strengthen our relations with our tribal nations. We also acknowledge that words are not enough. We must ensure that our institution provides support, resources, and programs that increase access to all aspects of higher education for our American Indian students, staff, faculty, and community members.

Methods of Instruction and Work Expectations

Course Workload Expectations

The RR-BIG summer workshop combines different sessions including lectures, laboratory practice, seminars, roundtable discussions, and mentored group projects. On each day, there will be a 1.5hr lecture in the morning and a 1hr lecture and 1hr laboratory practice in the afternoon. Two roundtable discussions and one seminar will be held at noon each week. Small research groups will be formed by the end of the second week with a mentor assigned for each team. The mentored group projects will start in Week 3 and are expected to be completed within 3 months after the end of the summer course. An online webinar will be arranged for final presentations.

- The lectures will (1) teach fundamentals in neuroimaging research and selecting clinically meaningful mental health phenotypes; (2) tutor basic principles in genetics and statistical methods for imaging genomics analysis (e.g. heritability); and (3) guide participants to conduct an integrative neuroimaging genomics analysis from study design, to data manipulation and analysis, to result dissemination.
- The laboratory practice parallels the lectures and provides trainees hands-on tutoring to access and retrieve the neuroimaging, genomics and clinical data available in the ABCD database. The labs will provide practice in conducting exploratory and formal analysis to identify genetic variants associated with neuroimaging using appropriate statistical tools in the R platform.
- Seminars and roundtable discussions incorporate leading faculty sharing their research on the cutting edges of brain imaging genomics studies and experience for robust and reproducible research.
- The team projects enable trainees to apply their training to real brain imaging genomics research from study design to analysis result presentation, benefiting from complementary expertise of team members and professional guidance of faculty mentors.

Technology

You will use the following technology tools in this course. Please make yourself familiar with them.

- [R: The R Project for Statistical Computing](#)
- [GitHub](#)

Learning Community

School of Public Health courses ask students to discuss frameworks, theory, policy, and more, often in the context of past and current events and policy debates. Many of our courses also ask students to work in teams or discussion groups. We do not come to our courses with identical backgrounds and experiences and building on what we already know about collaborating, listening, and engaging is critical to successful professional, academic, and scientific engagement with topics.

In this course, students are expected to engage with each other in respectful and thoughtful ways.

In group work, this can mean:

- Setting expectations with your groups about communication and response time during the first week of the semester (or as soon as groups are assigned) and contacting the TA or instructor if scheduling problems cannot be overcome.
- Setting clear deadlines and holding yourself and each other accountable.
- Determining the roles group members need to fulfill to successfully complete the project on time.
- Developing a rapport prior to beginning the project (what prior experience are you bringing to the project, what are your strengths as they apply to the project, what do you like to work on?)

In group discussion, this can mean:

- Respecting the identities and experiences of your classmates.
- Avoid broad statements and generalizations. Group discussions are another form of academic communication and responses to instructor questions in a group discussion are evaluated. Apply the same rigor to crafting discussion posts as you would for a paper.
- Consider your tone and language, especially when communicating in text format, as the lack of other cues can lead to misinterpretation.

Like other work in the course, all student to student communication is covered by the Student Conduct Code.

Use of AI

The use of artificial intelligence tools (including but not limited to ChatGPT, Grammarly, image generators, or code-generating software) is not permitted in this course unless explicitly authorized by the instructor for a specific assignment. All submitted work must be your own and reflect your individual understanding and effort. Use of AI tools without permission will be considered a violation of the course's academic integrity policy and may result in disciplinary action in accordance with institutional guidelines.

Course Text & Readings

Textbook

There is no textbook required for this class.

Additional Resources (Optional):

Ziegler, Andreas, Inke R. König, and Friedrich Pahlke. A Statistical Approach to Genetic Epidemiology: Concepts and Applications, with an E-learning platform. John Wiley & Sons, 2010.

Mills, Melinda C., Nicola Barban, and Felix C. Tropf. An introduction to statistical genetic data analysis. MIT Press, 2020.

Andrea S. Foulkes. Applied Statistical Genetics with R: For Population-based Association Studies (Use R!) Springer, 2009.

Journal Articles

This course uses journal articles, which are available via the University Libraries' E-Reserves and will be linked from the canvas course site. It is good practice to use a citation manager to keep track of your readings. More information about citation managers is available at [Citation management | University of Minnesota Libraries](#).

Course Outline/Weekly Schedule

This course has specific deadlines. All coursework must be submitted via the course site before the date and time specified on the site. **Note: assignments are due by 11:59pm Central Time unless indicated otherwise.**

Week 1: Basics of Brain Anatomy and Neuroimaging Data, July 13 to July 17, 2026

Lectures (9:00 - 10:30 a.m. ; 1 - 2 p.m.)

- Day 1: Introduce scientific questions in brain genomics and use of neuroimaging as intermediate phenotypes
- Day 2: Overview of brain structure and parcellation
- Day 3: Introduce neuroimaging terminology (voxel, ROI) and modalities (structural and functional MRI)
- Day 4: Extraction of meaningful phenotypes
- Day 5: Reproducible research

Seminars/Roundtable discussions (11:30 a.m. - 12:30 p.m.)

- Day 1: Big Data vs. Deep Phenotyping, perspectives towards 21st century neuroimaging research
- Day 3: Progress towards brain imaging data standardization and harmonization
- Day 5: Pitfalls and considerations for MRI phenotypes

Lab Activities (2:30 p.m. - 3:30 p.m.)

- Day 1: Accessing (MSI Sandbox/NDA LASSO) ABCD database
- Day 2: Data standards (BIDS), processing (fmriprep) and curation (CuBIDS) workflows
- Day 3: Viewing (workbench) neuroimaging data

- Day 4: Morphometry data extraction (FreeSurfer) and QC (BrainSwipes)
- Day 5: Reproducible Coding Practices on the LASSO Sandbox (spec-kit)

Week 2: Genomics Data and Methods for Heritability and Polygenic Risk Score Analysis, July 20 to July 24, 2026

Lectures (9:00 - 10:30 a.m. ; 1 - 2 p.m.)

- Day 1: Genetic data foundations for brain imaging genetics, genotype data formats, QC concepts, and statistical genetics motivation
- Day 2: Relatedness, kinship estimation, mixed effect models, and genomic relationship matrices
- Day 3: Population stratification, ancestry inference, admixture, and nonlinear methods including PCA, UMAP, and VAE
- Day 4: Heritability estimation in family-based and SNP-based settings, with attention to multi-site and multi-ancestry studies
- Day 5: Polygenic risk score analysis, evaluation, portability, and statistical and machine learning approaches in diverse samples

Seminars/Roundtable discussions (11:30 a.m. - 12:30 p.m.)

- Day 1: Mendelian Randomization
- Day 3: Interpretation, limitations, and responsible use of polygenic risk scores
- Day 5: Recent advances in polygenic risk scores

Lab Activities (2:30 p.m. - 3:30 p.m.)

- Day 1: Run QC pipeline and review genetic data types (.vcf.gz, .bed, .bim, .fam, .ped)
- Day 2: Explore relatedness and kinship outputs and identify duplicates, siblings, and cryptic relatives
- Day 3: Explore ancestry outputs and compare PCA, UMAP, and VAE on simulated admixed data
- Day 4: Run SNP heritability analyses and compare estimates across ancestries, sites, and estimators
- Day 5: Run PRS workflow and explore outputs, evaluation metrics, and portability across samples

Week 3: Study Design and Integrative Analysis of Brain Imaging Genomics, July 27 to July 31, 2026

Lectures (9:00 - 10:30 a.m. ; 1 - 2 p.m.)

- Day 1: Design a neuroimaging genetics study (generate hypothesis, select neuroimaging phenotypes and type of study, sample size/power calculation)

- Day 2: Data curation and exploration (inconsistency identification/harmonization)
- Day 3: Heritability and GWAS analysis of neuroimaging phenotypes and challenges
- Day 4: PRS analysis and model assessment/validation
- Day 5: Result presentation and documentation

Seminars/Roundtable discussions (11:30 a.m. - 12:30 p.m.)

- Day 1: Evaluating General Population PRS in Childhood Cancer Survivors for anthropometric traits: Insights from Variance Decomposition Due to Gene-by-Treatment Interaction
- Day 3: Integrative Bayesian Methods for imaging-genomic data
- Day 5: Considerations for reproducible and robust research

Lab Activities (2:30 p.m. - 3:30 p.m.)

- Day 1: Phenotype selection and study design (sample size/power calculation)
- Day 2: Data cleaning, covariate selection, neuroCombat, and quality control process
- Day 3: Heritability (precomputed) and GWAS analysis for selected phenotypes
- Day 4: PRS model development, testing, and validation
 - Caution: doesn't differentiate between genetic correlation and environmental correlation of phenotypes
- Day 5: Visualization, presentation, and documentation

Webinar (Monday, August 31)

- Group project presentation

Assessments

Student performance in this course will be evaluated based on the successful completion of lab assignments and a final project.

Lab assignments will be assigned at the end of lab sessions every day and are expected to be completed before the lab session of the following lecture day. They are designed to reinforce course concepts through hands-on practice and account for 50% of the final course grade.

Group projects will be initiated in the second week of the course and continued and completed after the in-person classes via remote group study and mentoring. Group members are expected to meet with their mentors at least once a week to discuss the progress of the project. Two webinar sessions will be held one month after the end of the summer course for the students to present their group project. The final group projects will account for the remaining 50% of the final grade and will assess students' ability to integrate and apply course concepts in a comprehensive manner.

Both components must be completed in order to earn a passing grade in the course. Extensions on lab assignments or the final project may be granted at the instructor's discretion when students

communicate in advance and provide a reasonable justification. Requests for extensions should be made as early as possible, and approval is not guaranteed.

Evaluation & Grading

Course evaluation will be based on completion of lab assignments (50%) and a final project (50%). A letter grade will be determined from the percentage of points each student receives. The curve for final grades will be: A = 95-100; A- = 90-94; B+ = 85-89; B = 80-84; B- = 75-79; C+ = 70-74; C = 65-69; C- = 60-64; F = below 60. For those registered S/N, S = 60-100.

Grading Scale

The University uses plus and minus grading on a 4.000 cumulative grade point scale in accordance with the following, and you can expect the grade lines to be drawn as follows:

% In Class	Grade	GPA
93 - 100%	A	4.000
90 - 92%	A-	3.667
87 - 89%	B+	3.333
83 - 86%	B	3.000
80 - 82%	B-	2.667
77 - 79%	C+	2.333
73 - 76%	C	2.000
70 - 72%	C-	1.667
67 - 69%	D+	1.333
63 - 66%	D	1.000
< 62%	F	0.000

- A = achievement that is outstanding relative to the level necessary to meet course requirements.

- B = achievement that is significantly above the level necessary to meet course requirements.
- C = achievement that meets the course requirements in every respect.
- D = achievement that is worthy of credit even though it fails to meet fully the course requirements.
- F = failure because work was either (1) completed but at a level of achievement that is not worthy of credit or (2) was not completed and there was no agreement between the instructor and the student that the student would be awarded an I (Incomplete).
- S = achievement that is satisfactory, which is equivalent to a C- or better
- N = achievement that is not satisfactory and signifies that the work was either 1) completed but at a level that is not worthy of credit, or 2) not completed and there was no agreement between the instructor and student that the student would receive an I (Incomplete).

Missed Exams and Late Assignments

Policy for making up missed exams and grading late work, consistent with Administrative Policy: [Excused Absences and Makeup Work](#).

Attendance Requirements

Regular attendance and active participation are required, as lab activities and instruction are conducted during scheduled class meetings and cannot always be replicated outside of class.

Saving & Submitting Coursework

- Documents that students submit are considered final; students may not submit more than one version or draft of each assignment.
- Assignments should be submitted via Canvas; emailed documents will not be accepted.

Technical Issues with Course Materials

You are expected to submit all coursework on time and it is your responsibility to ensure that your work is submitted properly before the deadline.

If you experience technical difficulties while navigating through the course site or attempting to submit coursework:

- Go to [Quick Help](#).
- Connect with the appropriate person or office within 30 minutes of the problem's occurrence.
 - Provide as much information as possible, so the tech team can best help you as soon as possible.
 - You can expect a response within 1-2 business days to help resolve the problem.

SPH and University Policies & Resources

Course policies

Grading and grade definitions

The University has two distinct grading scales: A-F and S-N. The Grading and Transcripts policy articulates the scales and definitions for these scales, as well as other transcript notations.

For more information: [Grading and Transcripts: Crookston, Morris, Rochester, Twin Cities](#)

Scholastic Dishonesty, Plagiarism, Cheating, etc.

As students in a university community, you are expected to do your own academic work, and to cite sources of knowledge that you draw on in completing your assignments and other academic work. Failing to do so is scholastic dishonesty, which includes: plagiarism; cheating on assignments or examinations, including the unauthorized use of online learning support and testing platforms; engaging in unauthorized collaboration on academic work, including the posting of student-generated coursework on online learning support and testing platforms not approved for the specific course in question; taking, acquiring, or using course materials without faculty permission, including the posting of faculty-provided course materials on online learning support and testing platforms; submitting false or incomplete records of academic achievement; acting alone or in cooperation with another to falsify records or to obtain dishonestly grades, honors, awards, or professional endorsement; altering, forging, misrepresenting, or misusing a University academic record; or fabricating or falsifying data, research procedures, or data analysis. When it is determined that a student has cheated, the student may be given an "F" or an "N" for the course, and may face additional sanctions from the University. For additional information, please see: Teaching and Learning: Instructor and Unit Responsibilities.

Additional resources pertaining to academic integrity can be found through the [Office for Community Standards](#) and the [Center for Educational Innovation](#).

Beware of websites that advertise themselves as being "tutoring websites." It is not permissible to upload any instructor materials to these sites without their permission or copy material for your own homework assignments from these various sites. When you have additional questions and concerns, please speak with or email your instructor to clarify practices expected for the course. Your instructor can respond to your specific questions regarding what would constitute scholastic dishonesty in the context of a particular class, for example, when and whether collaboration on assignments is permitted, when citing sources is required and what citation method to use, or when and which electronic aids are permitted or prohibited during an exam.

For more information, please see the related policies:

- [Board of Regents Policy: Student Conduct Code](#)
- [Administrative Policy: Teaching and Learning: Instructor and Unit Responsibilities \(Crookston, Morris, Rochester, Twin Cities\)](#)

Excused Absences and Makeup Work

Students will not be penalized for absence during the semester due to excused circumstances. Such circumstances include verified illness, participation in intercollegiate athletic events, subpoenas, jury duty, military service, bereavement, and religious observances. Such circumstances do not include voting in local, state, or national elections.

For complete information, please see Administrative Policy: [Excused Absences and Makeup Work: Crookston, Morris, Rochester, Twin Cities](#).

Student Conduct

The University seeks an environment that promotes academic achievement and integrity, that is protective of free inquiry, and that serves the educational mission of the University. Similarly, the University seeks a community that is free from violence, threats, and intimidation; that is respectful of the rights, opportunities, and welfare of students, faculty, staff, and guests of the University; and that does not threaten the physical or mental health or safety of members of the University community.

As a student at the University you are expected to adhere to the [Board of Regents Student Conduct Code](#). Note that the conduct code specifically addresses disruptive classroom conduct, saying “Disruption of the academic environment means engaging in behavior that substantially or repeatedly interrupts either the instructor’s ability to teach and/or a student’s ability to learn.”

For more information, please see these related policies:

- Board of Regents Policy: [Student Conduct Code](#)
- Administrative Policy: [Teaching and Learning: Student Responsibilities \(Crookston, Morris, Rochester, Twin Cities\)](#)

Sexual Harassment, Sexual Assault, Stalking and Relationship Violence

The University prohibits sexual misconduct, and encourages anyone experiencing sexual misconduct to access resources for personal support and reporting. If you want to speak confidentially with someone about an experience of sexual misconduct, please contact a confidential resource on your campus. If you want to report sexual misconduct, or have questions about the University’s policies and procedures related to sexual misconduct, please contact your campus [Title IX office or relevant policy contacts](#).

Instructors are required to share information they learn about possible sexual misconduct with the campus Title IX office that addresses these concerns. This allows a Title IX staff member to reach out to those who have experienced sexual misconduct to provide information about personal support resources and options for investigation. You may talk to instructors about concerns related to sexual misconduct, and they will provide support and keep the information you share private to the extent possible given their University role.

For more information, please see Administrative Policy: [Sexual Harassment, Sexual Assault, Stalking and Relationship Violence](#).

Equity, Diversity, Inclusion, and Equal Opportunity

The University provides equal access to and opportunity in its programs and facilities, without regard to race, color, creed, religion, national origin, gender, age, marital status, familial status, disability, public assistance status, membership or activity in a local commission created for the purpose of dealing with discrimination, veteran status, sexual orientation, gender identity, or gender expression.

For more information, please consult Board of Regents Policy: [Diversity, Equity, Inclusion, and Equal Opportunity](#).

Discrimination

All University members are prohibited from engaging in, or assisting or abetting another's engagement in, discrimination and related retaliation (collectively "prohibited conduct"). The University of Minnesota (the "University") will take prompt and effective steps intended to end prohibited conduct; prevent its recurrence; and, as appropriate, remedy its effects. In order to foster an environment free from prohibited conduct, all University members are encouraged to take reasonable prudent actions to prevent, stop, and report all acts of prohibited conduct.

For more information, please see the Administrative Policy: [Discrimination](#).

Disability Accommodations

The University of Minnesota is committed to creating learning environments that are inclusive and accessible to all students. If you are experiencing disability-related barriers to learning in your courses, the Disability Resource Center (DRC) is the office that collaborates with students to explore reasonable accommodations, tools, and resources.

- If you are registered with the DRC and have a current accommodation letter please share your letter with your instructor as soon as possible so that you can discuss how your accommodations will be implemented in the course. The sooner your instructor knows about your disability access-needs, the more equipped they can be to facilitate accommodations. You may reach out to your instructor or your access consultant if you have any questions or concerns about your accommodations.
- If you are not registered with the DRC and are experiencing or think you may be experiencing disability related to a mental health, attention, learning, chronic health, sensory, or physical condition, and would like to discuss accommodations and/or resources, please contact the DRC on our campus (e.g., Twin Cities campus – 612.626.1333).
- If you have a short-term medical condition, such as a broken arm, your instructor may be able to assist in minimizing classroom barriers. In situations where additional assistance is needed, you should contact the DRC as noted above.

For additional information, please see the Board of Regents Policy: [Disability Resources](#)

Academic Freedom and Responsibility

Academic freedom is a cornerstone of the University. Within the scope and content of the course as defined by the instructor, it includes the freedom to discuss relevant matters in the classroom. Along with this freedom comes responsibility. Students are encouraged to develop the capacity for critical judgment and to engage in a sustained and independent search for truth. Students are free to take reasoned exception to the views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.*

Reports of concerns about academic freedom are taken seriously, and there are individuals and offices available for help. Contact the instructor, the Department Chair, your adviser, the associate dean of the college, or the Vice Provost for Faculty and Academic Affairs in the Office of the Provost.

* Language adapted from the American Association of University Professors "Joint Statement on Rights and Freedoms of Students."

For more information, please see the Board of Regents Policy: [Academic Freedom and Responsibility](#)

Course-related Resources and Information for Students

Mental Health and Stress Management

As a student you may experience a range of issues that can cause barriers to learning, such as strained relationships, increased anxiety, alcohol/drug problems, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may lead to diminished academic performance and may reduce your ability to participate in daily activities. University services are available to assist you. You can learn more about the broad range of confidential mental health services available on campus via the [Student Mental Health Website](#). As an instructor/University community member, we care about the wellbeing of students. If health, safety, or mental health concerns are conveyed, we may consult with campus support offices to provide support and resources to a student.

Use of Personal Electronic Devices in the Classroom

The University establishes the right of each faculty member to determine if and how personal electronic devices are allowed to be used in the classroom.

Related University policy: [Teaching and Learning: Student Responsibilities \(Crookston, Morris, Rochester, Twin Cities\)](#)

Appropriate Student Use of Class Notes and Course Materials

Students may not distribute instructor-provided notes or other course materials, except to other members of the same class, during the same term, or with the express (written) consent of the instructor. Instructors have the right to impose additional restrictions on course materials in accordance with copyright and intellectual property law and policy. Students may not engage in the widespread distribution or sale of transcript-like notes or notes that are close to verbatim records of a lecture or presentation.

Related University policy: [Teaching and Learning: Student Responsibilities \(Crookston, Morris, Rochester, Twin Cities\)](#)

Dates and Deadlines

The following sites offer valuable information about University and academic dates and deadlines:

- [One Stop Student Services Academic Calendar](#)
- [One Stop Student Services Refund and Drop/add deadlines](#)